



Tribhuvan University
Institute of Engineering, Pulchowk College
Pulchowk, Lalitpur

Groundwater Resources of Nepal

Presented By:

Sagar Gyawali (078BCE135)

Sagar Poudel (078BCE138)

Samikshya Pokhrel (078BCE143)

Sandeep Dharel (078 BCE146)

Sushil Duwal (078BCE181)

Presentation Contents:

- Overview of the ground water resources of Nepal
- Kathmandu valley aquifer
- Terai aquifer
- Critical challenges and threat
- Sustainable development possibilities and uses of aquifers in Nepal

Overview of the ground water resources of nepal

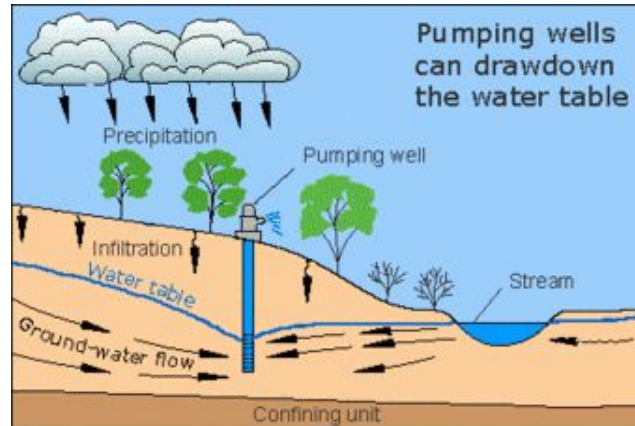


Source: GWRDB

- What is groundwater and why it matters in Nepal
- Hydrogeological regions (Hills, Terai, Kathmandu Valley)
- Main uses and value of groundwater in Nepal
- Current status: quantity & quality highlights
- Key challenges and governance actors
- Summary & references

What is Groundwater & Why it matters (quick background)

- Groundwater = water stored below ground in aquifers (saturated zone). It complements surface water.
- In Nepal, groundwater supports drinking water, irrigation, and industry, especially in Terai and urban centres.
- In hilly and mountainous areas, springs are often the primary local groundwater supply.



source:USGC.gov

Hydrogeological regions of Nepal

- **Terai and Inner Terai:** thick alluvial aquifers (sand, gravel) — high groundwater yield; widely used for irrigation and domestic water supply.
- **Hills and Mountains:** fractured bedrock and springs dominate; groundwater occurrence is more local and patchy.
- **Kathmandu Valley:** a bowl-shaped alluvial basin with shallow and deeper aquifers; intensive urban extraction has led to declining water tables and pollution risks.

Hydrogeological reasons and Use of Groundwater:



Source: wikipedia



Uses & socio-economic importance

- Domestic water supply in rural and urban areas (handpumps, tube wells, dug wells, boreholes).
- Irrigation in Terai: groundwater supports a significant fraction of agriculture — wells and shallow tube wells are common.
- Urban supply supplements (e.g., Kathmandu) where surface water and distribution networks are limited.

Groundwater quantity & recharge

- Recharge is driven by rainfall (highly seasonal) and stream–aquifer interactions; monsoon rains are key for recharge.
- Terai aquifers generally have high storage potential; Kathmandu's aquifer is limited and vulnerable to over-extraction.

Groundwater quality & health issues

- Contamination hotspots: microbiological contamination (coliforms) in many shallow sources, iron in Kathmandu, and **arsenic** occurrence in parts of the Terai.
- Urban pollution sources: leakage, inadequate sanitation systems, and industrial effluents that affect shallow wells.

Governance, data & institutions

- Key national bodies and resources: Ground Water Resources Development Board (GWRDB) — groundwater investigation and mapping;
- Kathmandu Valley Water Supply Management Board (KVWSMB) for urban water.

References :

Ground Water Resources Development Board (GWRDB). (n.d.). *Publications and hydrogeological studies.* Government of Nepal. [GWRDB](#)

Kathmandu Valley Water Supply Management Board (KVWSMB). (n.d.). *Groundwater in Kathmandu Valley.* [KVWSMB](#)

Japan International Cooperation Agency (JICA). (n.d.). *Basic design study on the Terai groundwater.* [JICA Report](#)

Pandey, V. P., & Kazama, F. (n.d.). *Groundwater in Kathmandu Valley: Status, challenges and opportunities.* [ResearchGate](#)

Thakur, J. K., et al. (n.d.). *Arsenic contamination of groundwater in Nepal: An overview.* Water. MDPI.

Winrock International, & United States Agency for International Development (USAID). (n.d.). *Nepal water resources profile.* Winrock International.

Presentation Contents:

- Overview of the ground water resources of Nepal
- Kathmandu valley aquifer
- Terai aquifer
- Critical challenges and threat
- Sustainable development possibilities and uses of aquifers in Nepal



Kathmandu Valley Aquifer: Aquifer: Lifeline Under Threat

Beneath the bustling streets of Kathmandu lies a critical, yet vulnerable, resource: its groundwater aquifer. This presentation explores the vital role of the aquifer, the challenges it faces, and the urgent need for sustainable management.

The Hidden Reservoir Beneath Kathmandu

The Kathmandu Valley is a geological marvel, cradling up to 600 meters of Quaternary fluvio-lacustrine sediments. These ancient layers form the basin's foundation, holding invaluable water resources.

Two distinct aquifer systems define this hidden reservoir:

- The **shallow unconfined aquifer**, typically found within 0-30 feet of the surface.
- The **deep confined aquifer**, extending to depths of 1000-1200 feet.

Together, these aquifers are the lifeblood for approximately 1.4 million residents, fulfilling their daily water requirements.



Groundwater Recharge: Seasonal Lifeblood



The dynamic interplay of seasonal rainfall and geographical features ensures the valley's aquifers are constantly replenished, though at varying rates.

Overextraction and Declining Water Tables



Since the 1970s, rapid urbanization has triggered a severe depletion of the Kathmandu Valley's precious aquifers. Unchecked growth has led to unsustainable water extraction.

- Projections indicate a significant drawdown by 2050: up to **7.4 meters** in the wet season and **5.3 meters** in the dry season within urban centers.
- The central urban areas and southern parts of the valley are experiencing the most drastic drops in groundwater levels, threatening long-term water security.



Contamination Crisis in Both Both Aquifers

Shallow Aquifer Contamination

This aquifer is severely impacted by **E. coli** and **nitrates**, primarily originating from inadequate sewage systems and unfiltered surface runoff.

Deep Aquifer Pollution

Even the deeper aquifer is not spared, showing alarming levels of **arsenic** and other **heavy metals**, posing significant public health risks.

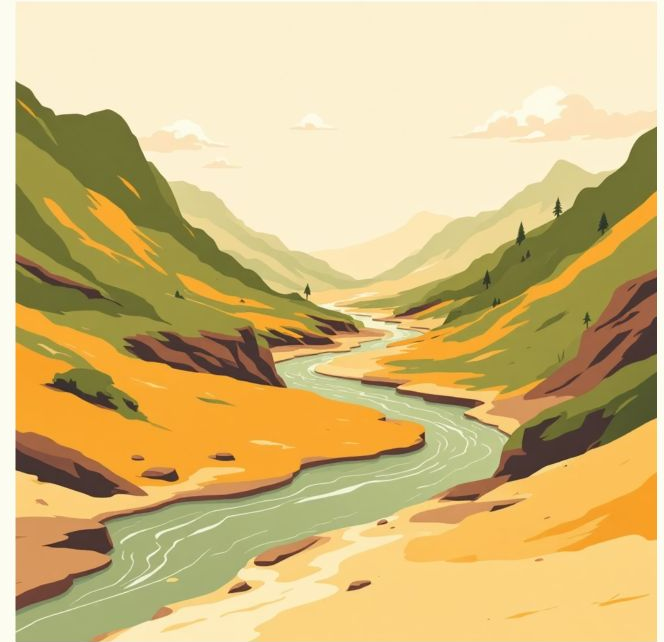
The pervasive contamination means tap water is largely unsafe for consumption, forcing residents to rely on expensive bottled mineral water as the norm.

Groundwater Flow and Seasonal Dynamics

The shallow groundwater system in Kathmandu exhibits dynamic fluctuations throughout the year. Levels can vary significantly, ranging from 0.6 meters to 13 meters depending on the season.

During the monsoon season, intensive rainfall leads to substantial recharge, causing the water table to rise by an average of approximately 35%. This influx is crucial for replenishing the aquifer after the dry months.

Groundwater naturally flows towards the valley's major river systems, such as the Bagmati River. This movement can lead to seepage and, in some areas, contributes to erosion risks along riverbanks.

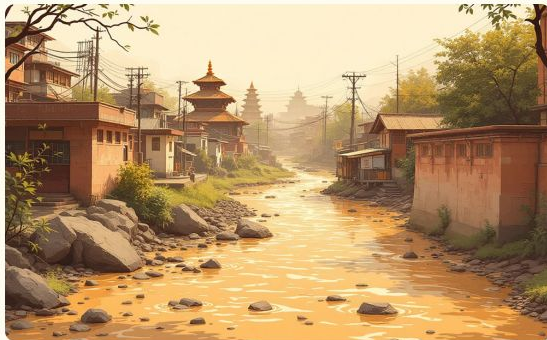


Infrastructure and Water Management Challenges



Limited Treatment Capacity

The Sundarijal treatment plant serves over 1 million people, but the valley lacks large-scale dams or canals for comprehensive water distribution.



Wastewater Crisis

Only 25% of wastewater receives treatment, leaving the Bagmati River heavily polluted and impacting groundwater quality.

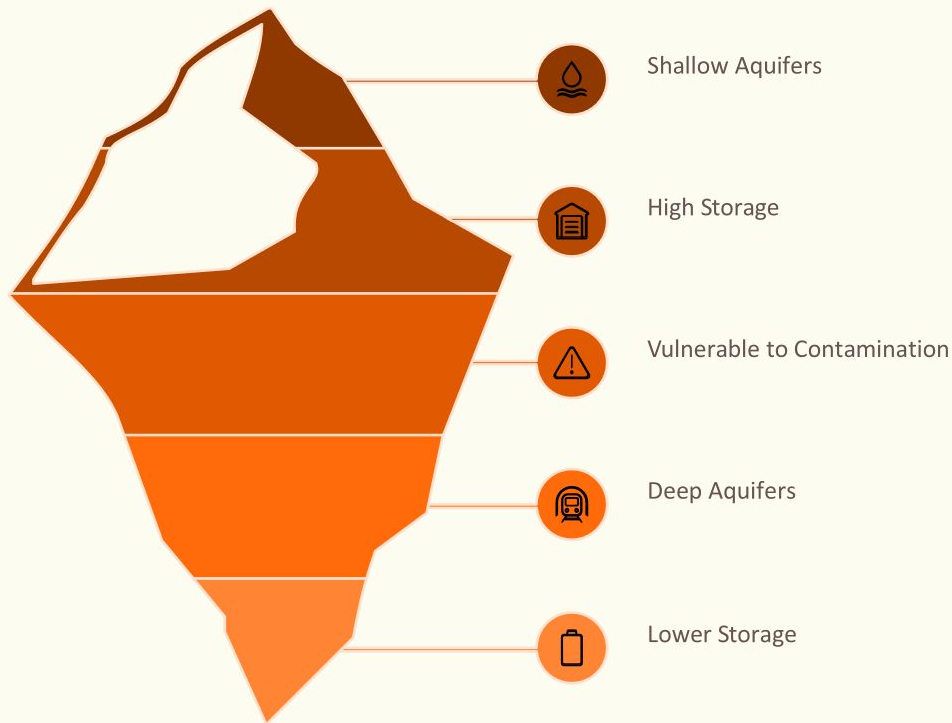


Management Structure

Kathmandu Upatyaka Khanepani Limited (KUKL) has managed the water supply since 2008 under a public-private partnership, striving to address these complex issues.

Mapping Groundwater Potential for Sustainable Use

Utilizing advanced Geographic Information System (GIS) based models is pivotal for understanding and sustainably managing the Kathmandu Valley's groundwater resources.



These models help identify zones with high groundwater availability and crucial recharge potential. While shallow aquifers offer higher storage, their vulnerability to contamination necessitates careful management. Strategic groundwater development must therefore balance extraction with critical recharge preservation efforts.

The Stakes: Water Security in a Growing City

Despite Nepal's abundant freshwater resources, Kathmandu faces an escalating water crisis, exacerbated by significant infrastructure gaps and widespread pollution.

Climate change further intensifies the challenge, leading to more erratic monsoon patterns and increased flooding, which in turn place additional stress on the already vulnerable aquifers.

Without immediate and decisive action, the continued depletion and contamination of the aquifer pose severe threats to public health and the long-term resilience of the urban environment.



Call to Action: Protecting Kathmandu's Aquifers for the Future



Enhance Recharge

Implement forest conservation and managed aquifer recharge projects to replenish groundwater naturally.



Upgrade Treatment

Invest in advanced wastewater treatment to drastically reduce contamination of shallow aquifers.



Integrated Management

Adopt a holistic approach, combining surface and groundwater management for sustainable use.



Community Engagement

Foster community awareness and advocate for policy reforms to safeguard this vital resource for generations to come.

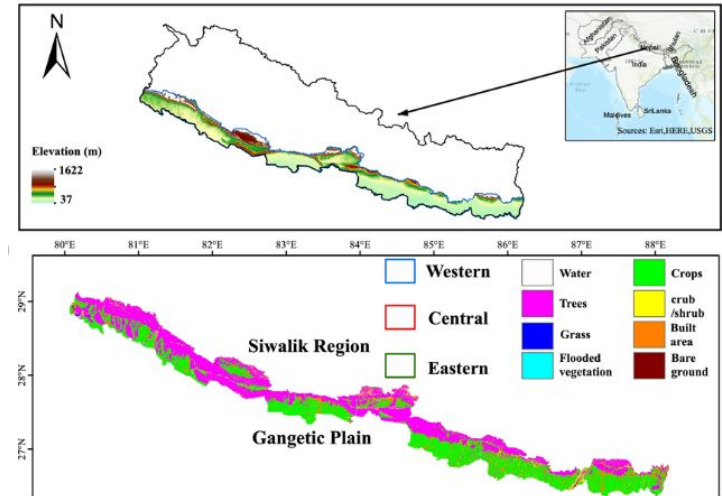
Presentation Contents:

- Overview of the ground water resources of Nepal
- Kathmandu valley aquifer
- Terai aquifer
- Critical challenges and threat
- Sustainable development possibilities and uses of aquifers in Nepal

Terai aquifer:

The terai aquifer is the part of Indo-Gangetic Basin (IGB), one of the world's largest groundwater reservoirs, extending across Nepal, India, Bangladesh, and Pakistan.

It can be considered as the groundwater power house of Nepal which contributes in the agriculture, household, drinking and many others.



[Geographical map depicting the Terai region of Nepal.](#)

Features of Terai aquifer:

- It is mainly composed unconsolidated alluvial deposits having the alternatives layers of sand, silt and clay.
- The thickness of terai aquifer increases from 50-100m near the foothills to 500m near some area of indian border.
- The terai aquifer is classified primarily unconfined to semi-confined as water is stored in between thick sand and gravel layer makes unconfined , while clay layer is also present which resist the flow of water makes confined.

Systematic hydrological Zone of terai(north to south)

Bhabar Zone

- **Location** : at the foothills of Shiwalik range.
 - **Materials** : consists of coarse, porous sediments like boulders, cobbles and pebbles.
 - **Infiltration capacity**: high infiltration capacity.
 - **Transmissivity** : very high transmissivity($> 5000 \text{ m}^2/\text{day}$) due to permeable sediments.
 - **Water table**: deep water table.
 - **Recharge mechanism**: Monsoon infiltration; subsurface inflow
- It acts as the primary natural recharge zone for the entire terai aquifer.

Systematic hydrological Zone of terai(north to south)

Central Terai

- **Location:** consists of middle terai districts mainly nawalparasi, kapilvastu, Rupandehi, Dang, Banke.
- **Materials :** thickest sand and gravel along with the presence of silt.
- **Infiltration capacity:** low infiltration capacity then bhabar zone but greater than southern terai.
- **Transmissivity:** high transmissivity of 2000-6000m²/day.
- **Recharge mechanism:** Lateral flow from Bhabar, river seepage, infiltration

Systematic hydrological Zone of terai(north to south)

Southern Terai

- **Location:** consists border area especially of eastern side.
- **materials:** finer sediments more silt and clay interlayers.
- **infiltration capacity:** low infiltration capacity than others due to presence of finer sediments.
- **transmissivity:** low transmissivity of 500-2000m²/day.
- **Water table:** shallow water table.
- **Recharge mechanism:** River losses/flooding, minor infiltration.

Systematic hydrological Zone of terai(north to south)

Journal of Nepal Hydrogeological Association (JNHA), Vol (2), (2025)

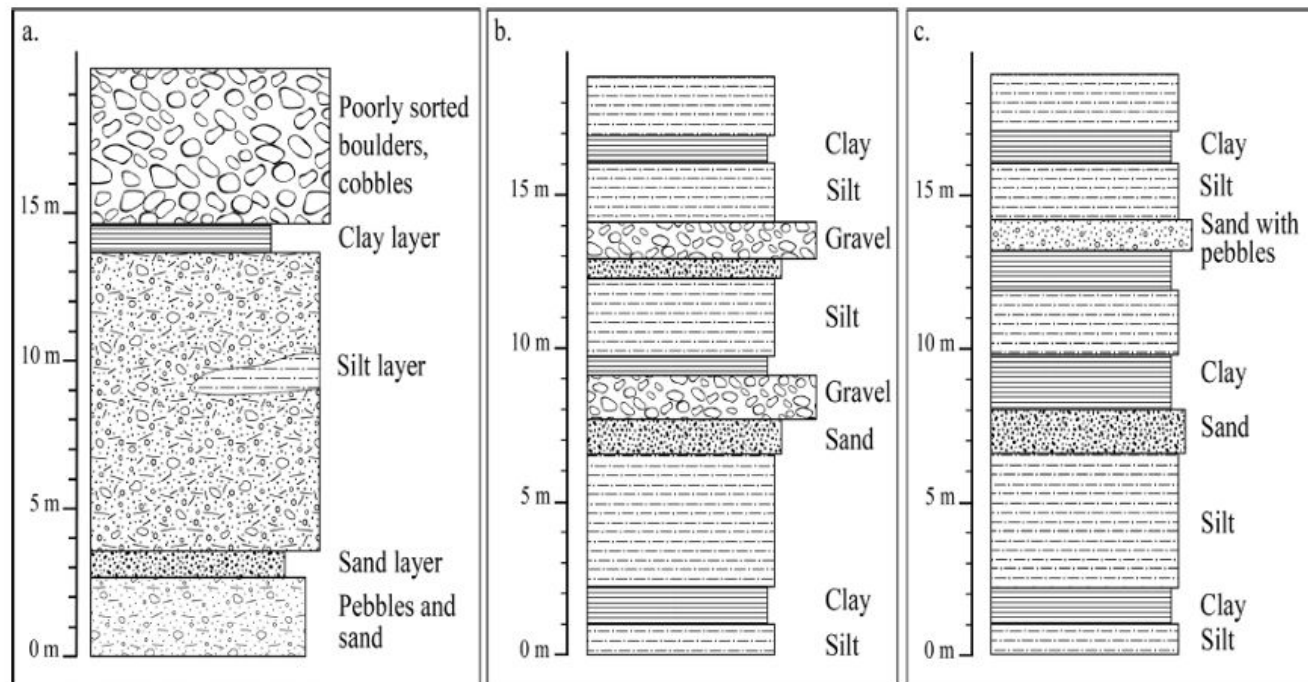
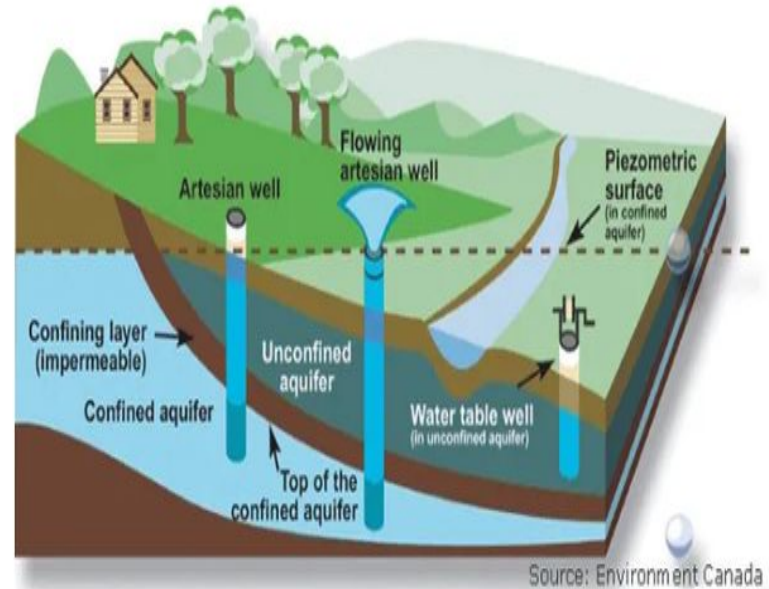


Fig. 2: Generalized lithologies of (a) Bhabar Zone, (b) Middle Terai, and (c) Southern Terai of Nepal (modified after Pathak, 2016)

Key roles of terai aquifer:

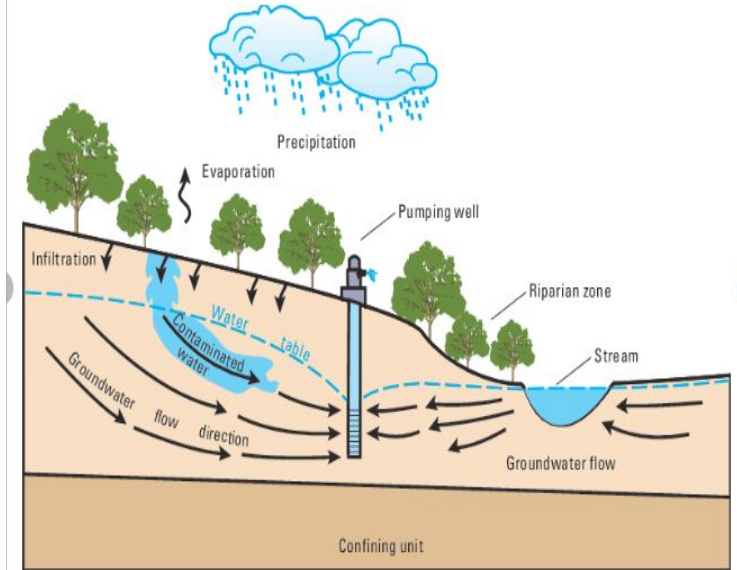
- **Drinking Water Supply:** The majority of the population in the Terai relies on groundwater as their primary source of drinking water. Groundwater is often more stable and requires less treatment than surface water as soil and rock layers within an aquifer act as natural filters.



[Live Science](#)
[Aquifers: Underground Stores of ...](#)

Key roles of terai aquifer:

- **Agricultural Production:** The Terai is also known agricultural heartland or the breadbasket of Nepal and the aquifer system is crucial for sustaining irrigation, especially during the dry season when surface water flows are low.

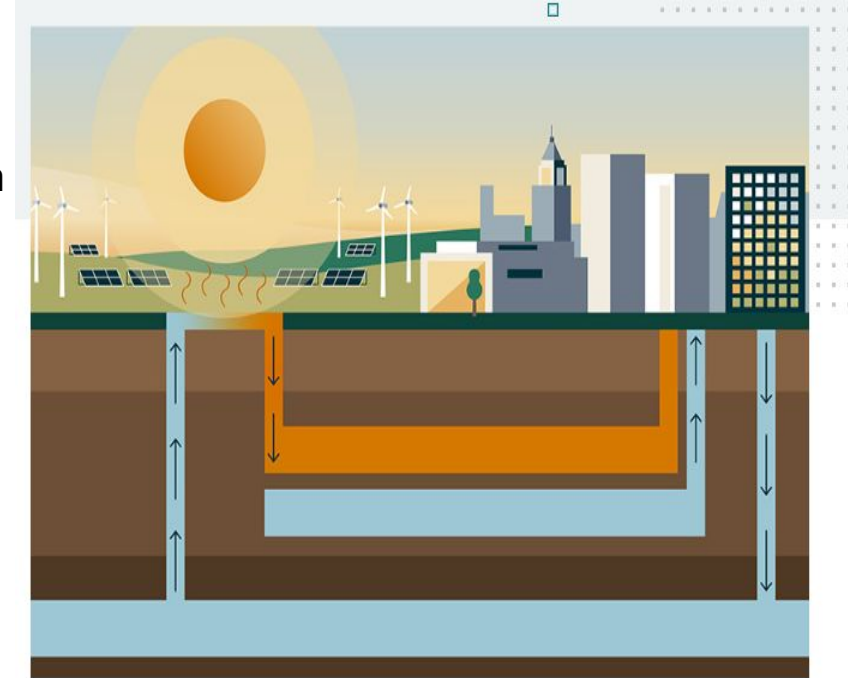


Schematic of a groundwater-flow system in an agricultural setting. In this example, infiltration of water from either precipitation or irrigation can transport phosphorus or other chemicals to the unsaturated zone and aquifer.

[Schematic of a groundwater-flow system in an agricultural setting.](#)

Key roles of terai aquifer:

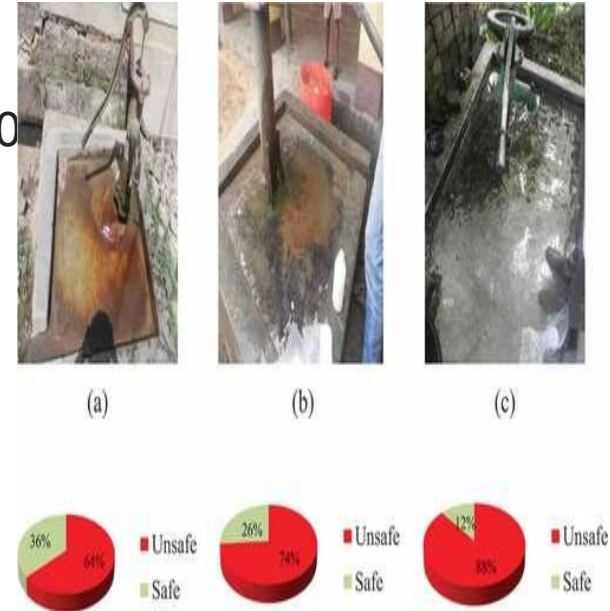
- **Industrial Use:** A significant amount of water is used in cooling systems to remove heat from equipment and for generating steam in boilers in industries. Underground Water Could be the Solution to Green Heating and Cooling.
- **Ecosystem support :** It provides a consistent base flow of water to rivers, lakes, and wetlands, especially during dry periods and droughts. This reliable water source is important for maintaining the biodiversity, and groundwater-dependent ecosystem.



Aquifer thermal energy storage (ATES) uses naturally occurring underground water to store energy that can be used to heat and cool buildings. (Credit: (Jenny Nuss/Berkeley Lab))

Groundwater contamination:

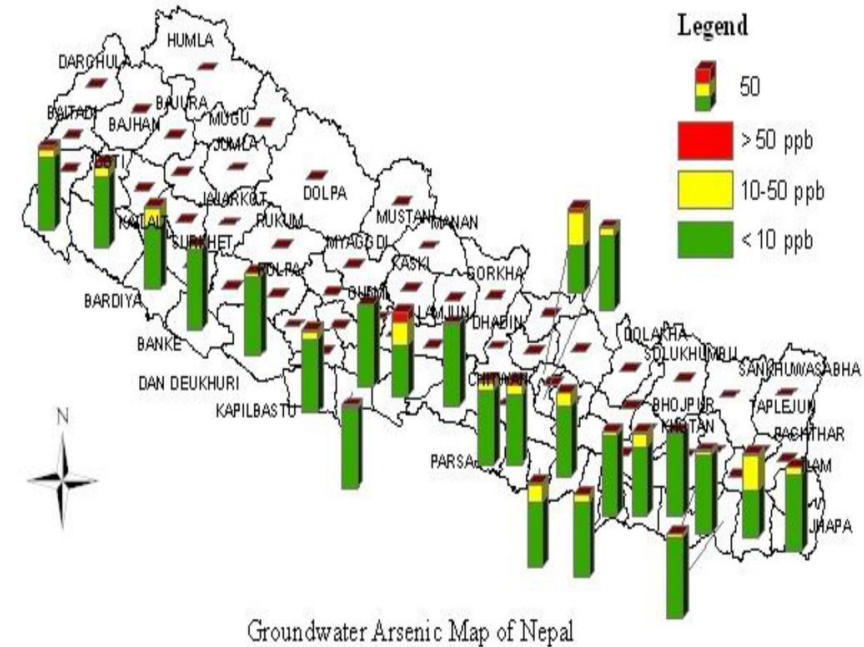
- **Arsenic:** In 1993, World Health Organization (WHO) had lowered the baseline of arsenic from 50 $\mu\text{g/L}$ to 10 $\mu\text{g/L}$ for “safe” limit for arsenic concentrations in drinking water, the reason for lowering the baseline limit was the widespread negative health effects on humans. 50 ppb is considered as acceptable limit in Nepal, advanced countries have already started to use the 10 ppb as the standard marker for drinking water.



[Full article: Tube well platform colour, as a proxy for identification of arsenic contaminated wells - case study from, alluvial aquifers of eastern India](#)

Groundwater contamination:

- **Arsenic:** Arsenic contamination in the Terai is widely researched in Nepal and according to Shrestha, et al. 2004, 7.4% of samples had values above 50 ppb whereas 23.7% had values above 10 ppb. measurements. study showed that arsenic levels usually decrease as the depth of the tube well increases, but tend to be higher in older tube wells (Shrestha et al., 2003). Even though only a small number of arsenic tests were carried out, deeper tube wells were found to have arsenic levels below 10 $\mu\text{g/L}$ (Smedley, 2005).



Groundwater contamination:

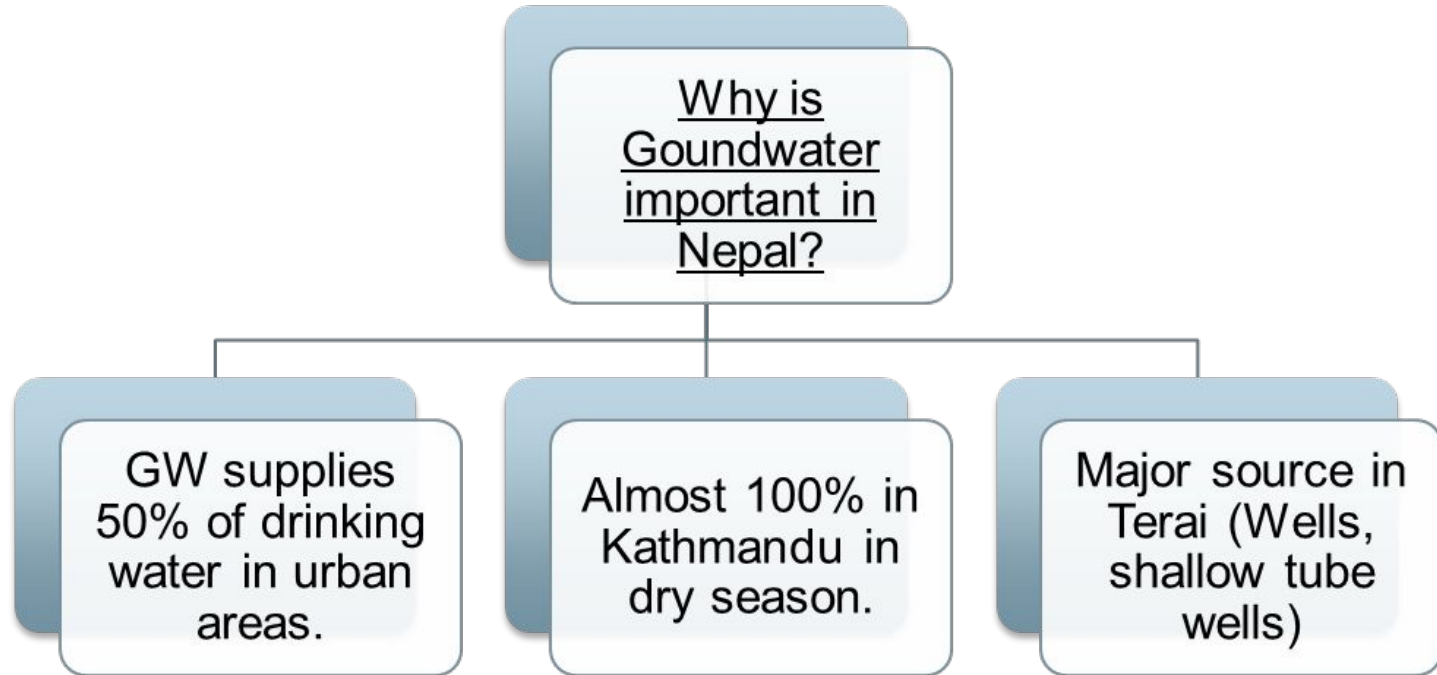
- Nitrate : Nitrate contamination is a significant and growing problem in the shallow aquifers of the Terai region due to anthropogenic activities such as extensive use of fertilizers and inadequate sewage/wastewater management. Deep aquifers in the region are generally less affected and naturally have low nitrate levels.
- Iron and manganese : Iron and manganese in Terai groundwater come naturally from underground rocks and soil. In areas where water has little oxygen, these metals easily dissolve and mix into the water supply.

References :

1. Pandey, N., Shrestha, B., Shrestha, A., Pandey, S., Pandey, N., & Sharma, G. (2025). *Mapping Aquifer Zones in the Terai Region of Nepal Using the Natural Electromagnetic Field Technique: A Case Study from Rajbiraj Municipality, Saptari*. **Journal of Nepal Hydrogeological Association**, 2(1), 127–143. <https://doi.org/10.3126/jnha.v2i1.85392>
2. Pokhrel, D., Bhandari, B. S., & Viraraghavan, T. (2009). *Arsenic contamination of groundwater in the Terai region of Nepal: An overview of health concerns and treatment options*. **Environment International**, 35(1), 157–161. <https://doi.org/10.1016/j.envint.2008.06.003>
3. Japan International Cooperation Agency. (Year). *Appendix-7 hydro-geology and groundwater development* (Feasibility Study on the Sunsari River Irrigation Project, Appendix-7) [PDF]. JICA. https://openjicareport.jica.go.jp/pdf/11716495_04.pdf
4. Mishra, K. K., Rauniyar, R. K., Jha, S. S., & Panthi, Y. (2025, August). *Understanding the ground water availability of Terai region in Nepal: Availability, impact of drought and recharge problems*. <https://doi.org/10.13140/RG.2.2.11530.17602>
5. Water Security in Nepal – Analysis of the Groundwater Resource and Governance
6. [Atomic Absorption Spectrophotometry: Principle, Parts, Uses](#)

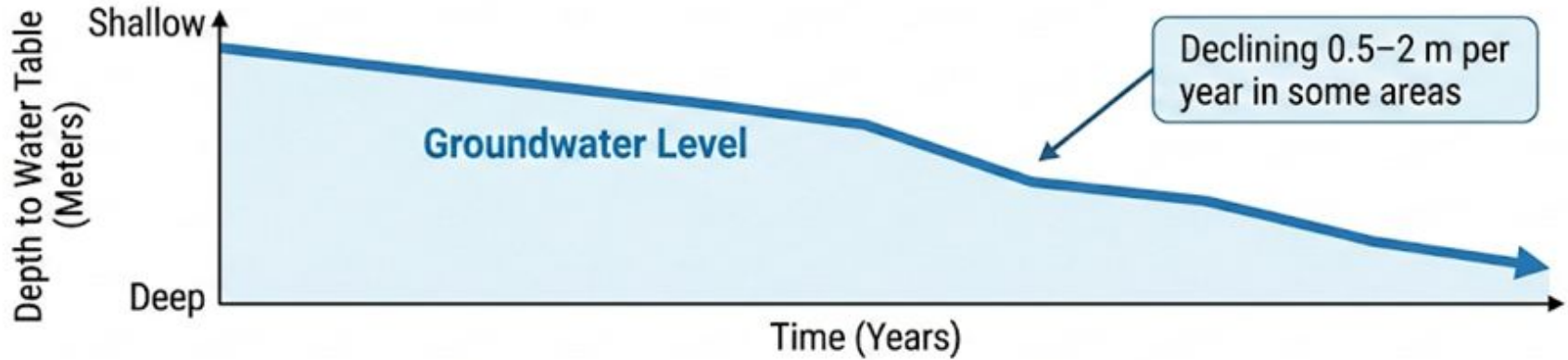
- Overview of the ground water resources of nepal
- Kathmandu valley aquifer
- Terai aquifer
- Critical challenges and threats
- Sustainable development possibilities and uses of aquifers in Nepal

Critical Challenges and threats



Over-extraction

Withdrawal > Natural recharge



Kathmandu Valley

- Rapid urbanization → thousands of private deep tube wells
- Land subsidence risk increasing

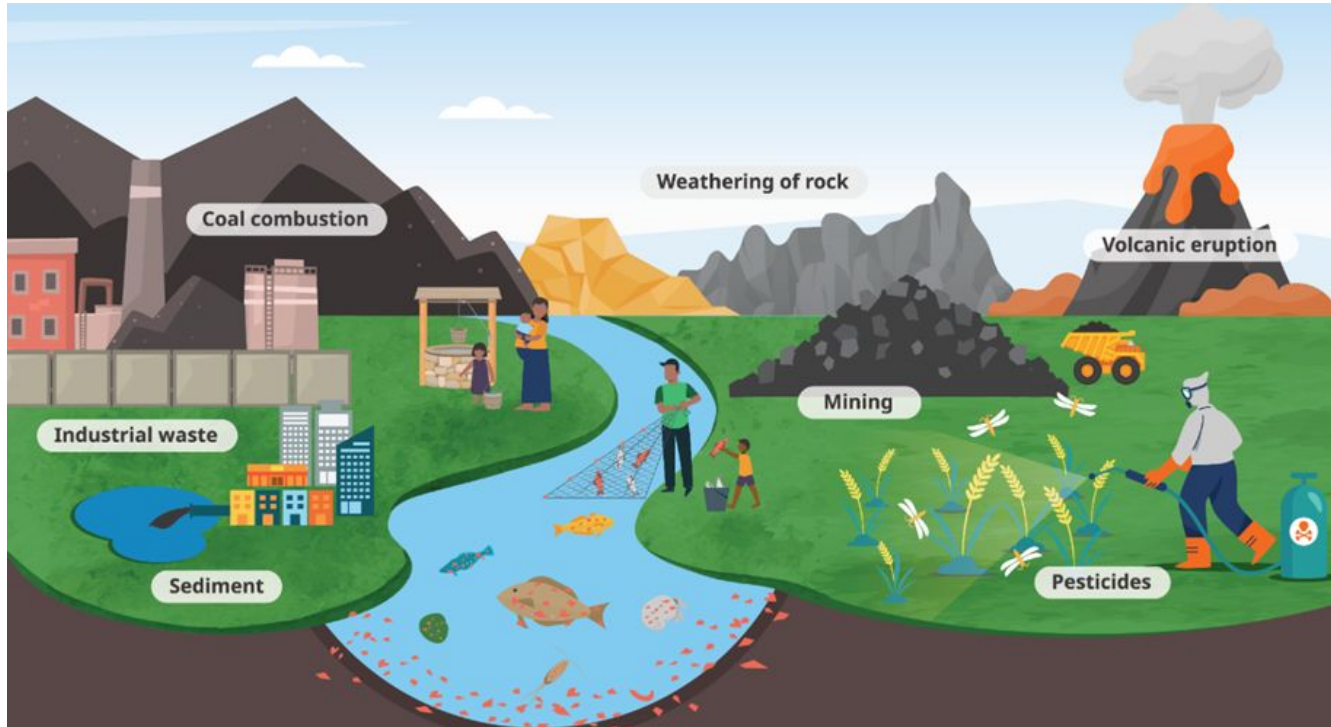


Terai

- Expansion of irrigation pumping
- Falling shallow aquifers in dry season

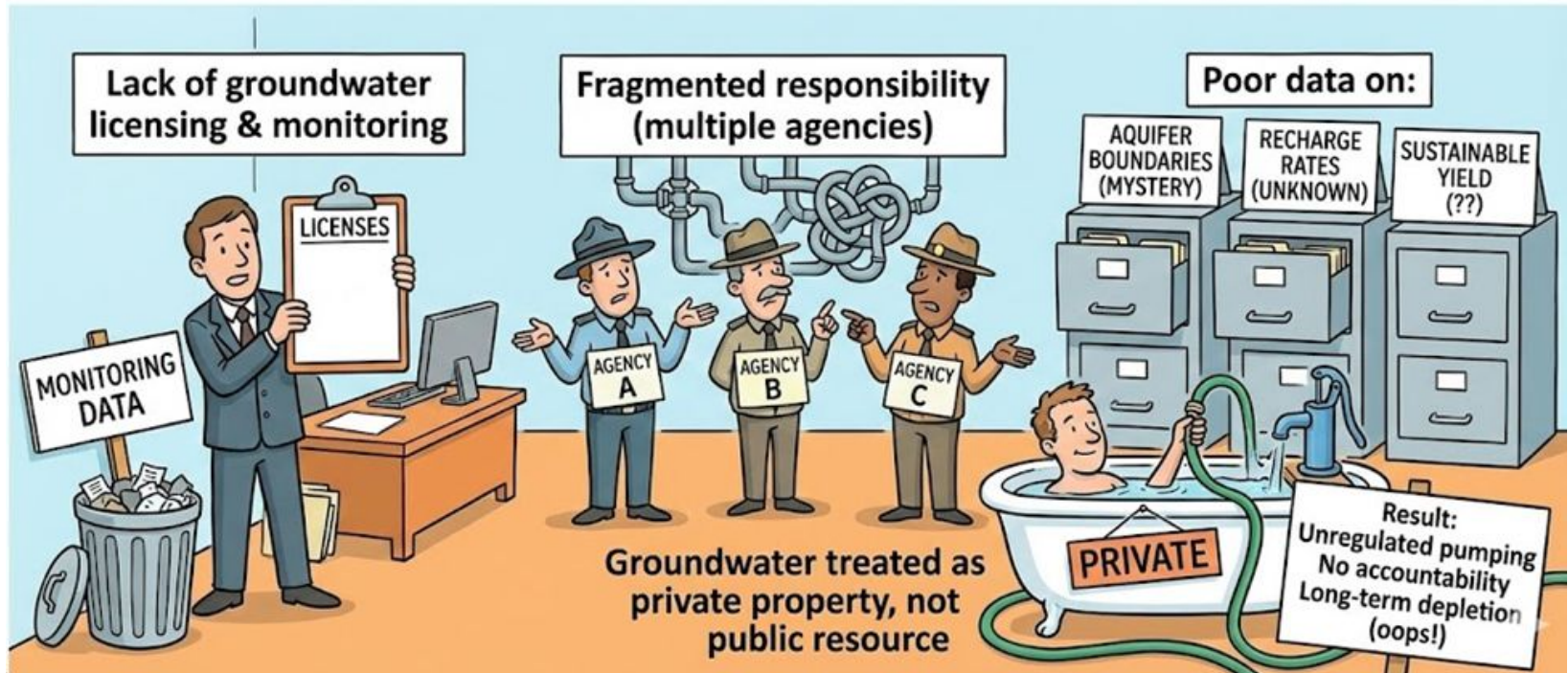
Groundwater Pollution

Pollutant like Nitrate, Arsenic, Pathogens, Iron etc



Weak Policies

Lack of GW licensing and monitoring, Poor data, No research.



Presentation Contents:

- Overview of the ground water resources of nepal
- Kathmandu valley aquifer
- Terai aquifer
- Critical challenges and threats
- Sustainable development possibilities and uses of aquifers in Nepal

Sustainable development possibilities and uses of aquifers in Nepal

1. Part of culture

- Groundwater aquifer of Nepal is mainly being used as a source of drinking water, for irrigation and industrial use.
- But, it also has a good potential for development of tourism in Nepal. Springs are a source of recreation and natural beauty thus, possessing the ability to attract local and international tourists.
- Ultimately, it helps to enhance the national economy.



Narchyang's Waterfall, Tatopani Myagdi

Source: www.theoutbound.com

Sustainable development possibilities and uses of aquifers in Nepal

2. Resilient seasonal water bearing reservoir

- As a result of high rainfall and presence of coarse-grained sediments in the northern part of Kathmandu Valley, it has high recharge potentials. Therefore, the aquifers in these regions can act as seasonal reservoirs.
- Their major advantages over surface reservoirs includes less vulnerability to evaporation and less contamination due to climate extremes.

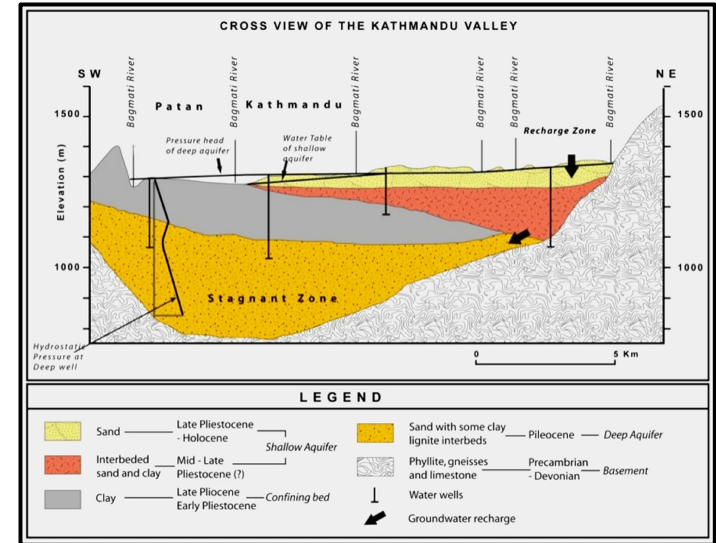


Fig: Kathmandu Valley aquifers

Source: link.springer.com

Sustainable development possibilities and uses of aquifers in Nepal

3. Expansion of irrigation in Terai for agricultural intensification

- As of 2022, only 35% of the total irrigation is contributed by groundwater through tube wells and springs.
- The irrigation in agricultural fields can be increased by installation of eco-friendly solar irrigation systems with provisions of recharge ponds and rain harvesting structures.
- Ultimately, the crop yield and productivity will be enhanced.

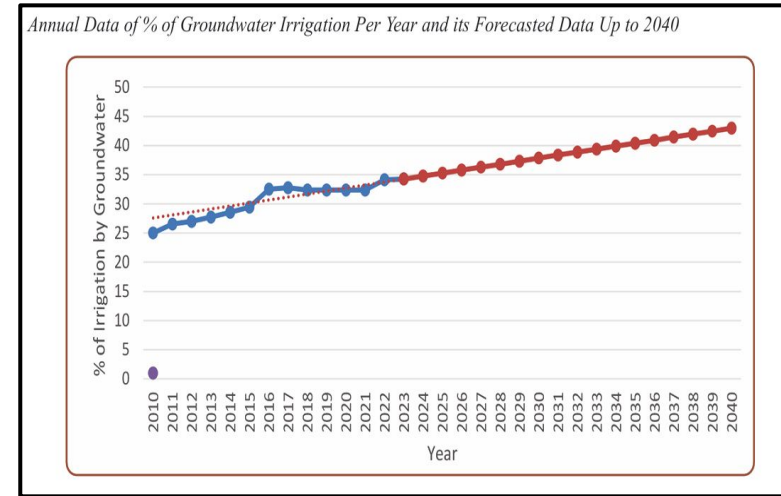


Fig: Irrigation by groundwater

Source: <https://doi.org/10.61916/prmn.2025.v04i01.006>

Sustainable development possibilities and uses of aquifers in Nepal

4. Conjunctive use of groundwater and surface water

- Due to increasing demand of water, it may be efficient to use the groundwater and surface water jointly for the optimal water supply.
- The best example of conjunctive use is seasonal switching i.e. using surface water in wet periods and groundwater in dry periods.
- The surplus water during wet periods can also be used for artificial recharge into the ground aquifer.

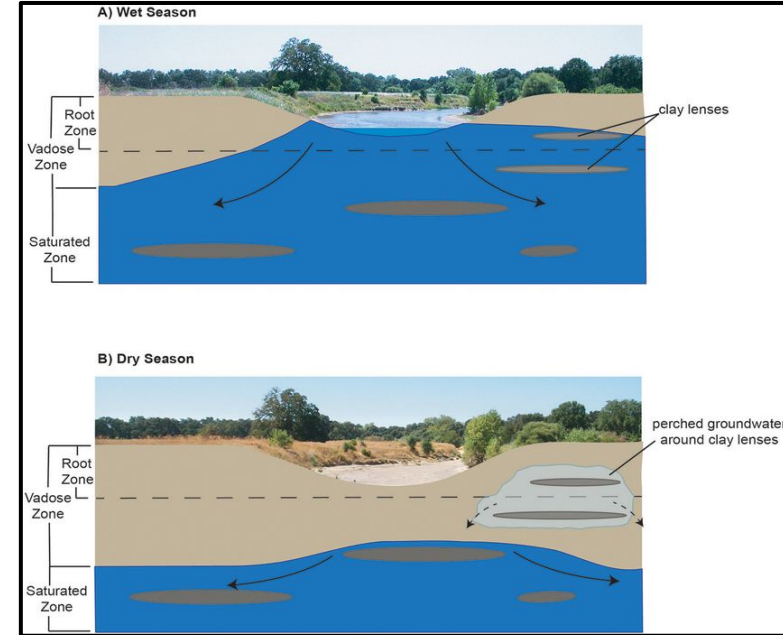


Fig: Seasonal Switching

Source: www.researchgate.net

References:

- V. P. Pandey *et al.*, **“Groundwater in Kathmandu Valley: Status, Challenges and Opportunities,”** *Kathmandu Valley Water Supply Management Board (KVWSMB)*, Nepal. ISBN: 1-4732-2
- M. M. Rohde, S. B. Sweet, C. Ulrich, and J. Howard, **“A Transdisciplinary Approach to Characterize Hydrological Controls on Groundwater-Dependent Ecosystem Health,”** *Front. Environ. Sci.*, vol. 7, Nov. 2019, doi: 10.3389/fenvs.2019.00175.
- R. Prajapati *et al.*, **“Streams, sewage, and shallow groundwater: stream-aquifer interactions in the Kathmandu Valley, Nepal,”** *Sustain. Water Resour. Manag.*, vol. 7, no. 5, Oct. 2021, doi: 10.1007/s40899-021-00542-8.
- S. Karki, **“Groundwater Sustainability and Solar Irrigation in Madhesh Province: Status, Challenges, and Integrated Solutions,”** *Int. Res. J. Parroha Mult. Campus*, vol. 4, no. 1, pp. 59–66, 2025, doi: 10.61916/prmn.2025.v04i01.006.

Past questions from this chapter :

- 2073 Bhadra (Regular)

9. Discuss the aquifer system of Kathmandu Valley. [4]

- 2072 Ashwin (Regular)

8. Discuss about aquifer system of Kathmandu Valley of Nepal. [6]

- 2072 Magh (Back)

9. Discuss about the groundwater development possibilities of Terai belt. [6]

- 2071 Bhadra (Regular/Back)

11. Discuss with the help of schematic zones the availability and possibilities of groundwater resources in Kathmandu Valley. [6]

THANK YOU

!!!